grifting by optimal transport

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Abstract

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1. Problem formulation (2026.05.15.)

"Drifting models" is a new (and somehow very popular) approach for generative modeling for learning to sample from a given target distribution ν_{tgt} accessible only via samples, given samples from a source distribution ν_{src} as input. Essentially, the idea behind drifting models is to define a Markov chain whose stationary distribution is equal to the target distribution, and directly estimate the stationary distribution of the proposed chain. The chain considered in the original work is a simple process that operates via rules of attraction and repellence: roughly speaking, a state is repelled by the source distribution and attracted by the target distribution. The original paper argues that this process converges to the target distribution under some conditions (once the specifics of the policy are set up correctly). Is there a good reason to believe that the proposed process is the best possible one for purposes of generative modeling? This is the question we ask in this note.

We thus formulate our problem as finding a control policy over the state space $\mathcal{X} = \mathbb{R}^d$ that yields ν_{tgt} as its stationary distribution. There are many possible policies that satisfy this, e.g., the process above or the one that draws states i.i.d. from the target distribution, and so on. We try to formulate criteria for a good policy in the language of infinite-horizon undiscounted optimal control below. The main idea is to find a policy with minimal *transient cost* incurred before the stationary distribution is reached. For concreteness, let $\pi : \mathcal{X} \to \Delta_{\mathcal{X}}$ be a stochastic policy and let us consider its stationary distribution ν^π (assuming that it exists). We denote measures on $\mathcal{X}$ by ν and the action of the transition kernel π is taking ν to the distribution $\pi\nu$ with density $(\pi\nu)(x) = \int_{\mathcal{X}} \pi(x|x^-)\nu(x^-)\mathrm{d}x^-$ (assuming everything has densities). Then, the transient state distribution induced by π (under initial-state distribution ν_{src}) is defined as

$$\begin{aligned} d^\pi &= \sum_{t=0}^{\infty} \left(\pi^t \nu_{\text{src}} - \nu^\pi\right) = \sum_{t=0}^{\infty} \pi^t \left(\nu_{\text{src}} - \nu^\pi\right) \\ &= \nu_{\text{src}} - \nu^\pi + \sum_{t=1}^{\infty} \pi^t \left(\nu_{\text{src}} - \nu^\pi\right) = \nu_{\text{src}} - \nu^\pi + \pi d^\pi. \end{aligned}$$

Well, the last part is not part of the definition but rather a consequence of the definition, which can be seen as the equivalent of the classic flow constraints for occupancy measures. We thus refer to d^π as the *transient occupancy measure* of π.

After reparametrizing the problem as in the classic LP formulations to eliminate the product πd^π, we thus define the state-next-state transient occupancy $\mu^\pi(x, x') = d^\pi(x)\pi(x'|x)$ instead and note that it satisfies the linear constraints

$$B\mu^\pi = \nu_{\text{src}} - \nu^\pi + F\mu^\pi,$$

where we have defined the forward and backward-marginalization operators F and B with their respective actions $(F\mu)(x) = \int_{\mathcal{X}} \mu(x', x)\mathrm{d}x'$ and $(B\mu)(x) = \int_{\mathcal{X}} \mu(x, x')\mathrm{d}x'$. Similarly, the stationary distribution ν^π

satisfies

$$\nu^\pi = \pi \nu^\pi.$$

This one still has a π in there which is no good. How can we reconcile this with the state-next-state occupancies? Those should obey the same conditional distribution π, which seems difficult to extract from μ while retaining the linearity of the constraints.

Here's an idea that might work. Since we hope to find a policy with stationary distribution equal to ν_{tgt}, we might as well replace the desired stationary distribution with this measure (and hope that the solution will somehow indeed yield a policy with this limiting measure). We thus simply consider the following linear program:

$$\begin{aligned} \min_{\mu} \int_{\mathcal{X}} \int_{\mathcal{X}} \mu(x, x') \left\| x - x' \right\|^2 \mathrm{d}x \mathrm{d}x' \\ \text{subject to} \qquad B\mu = \nu_{\text{src}} - \nu_{\text{tgt}} + F\mu. \end{aligned}$$

LOL this just looks like the classic Kantorovich LP all over again, except with the two constraints folded into one! Can this still be a good idea somehow? Let us consider a feasible point μ and try to understand it as a transient occupancy measure (denoting $\mu = \mu^\pi$ and π as the corresponding kernel):

$$d^\pi = \nu_{\text{src}} - \nu_{\text{tgt}} + \pi d^\pi.$$

Indeed, on the left-hand side we have $d^\pi(x) = \int_{\mathcal{X}} \mu^\pi(x, x') \mathrm{d}x'$ by definition, and on the right-hand side we have $\int_{\mathcal{X}} \mu^\pi(x, x') \mathrm{d}x = \int_{\mathcal{X}} d^\pi(x) \pi(x'|x) \mathrm{d}x$. Thus, we can iterate the above relation and get

$$d^\pi = \nu_{\text{src}} - \nu_{\text{tgt}} + \pi d^\pi = \nu_{\text{src}} - \nu_{\text{tgt}} + \pi(\nu_{\text{src}} - \nu_{\text{tgt}}) + \pi^2 d^\pi = \cdots = \sum_{t=0}^{\infty} \pi^t \left(\nu_{\text{src}} - \nu_{\text{tgt}}\right).$$

This sum is convergent if π has appropriate mixing properties. But notice that it does not imply that ν_{tgt} would be its stationary distribution. So we have basically proved that any quickly-mixing policy π is feasible, and that d^π corresponds to the *difference* of transient measures when started from distributions ν_{src} and ν_{tgt}.

So the problem remains: we should find a formulation that does actually yield ν_{tgt} as the stationary distribution. Maybe just explicitly require this via the constraint $F\mu = \nu_{\text{tgt}}$? [**Grg:** NO this is silly: μ is not directly related to the stationary distribution with equations like that.] This of course brings us back *exactly* to the Kantorovich LP (once the first constraint is simplified appropriately), which also verifies that the optimal policy should be such that it maps x deterministically to $M^\star(x)$ using the Monge map $M^\star$. But this doesn't mean that the above reformulation as a sequential decision-making problem is useless! We could in principle still use ideas from the VDT paper to solve this control problem via primal-dual methods: characterize the value functions by a variant of the Bellman equations (Poisson equations?), the optimal policy in terms of the gradient of the value function, and solve for the optimal value function by primal-dual methods. Maybe it could work, or maybe that's just nonsense... should be fun to think about it anyway.

1.1. Lazy approach 1: cost optimality under stationary distribution

The most straightforward way to fix up this formulation is to do away with the transient occupancies and simply work with stationary distributions. That is, simply consider the following adjustment to the classic undiscounted control LP:

$$\begin{aligned} \min_{\mu} \int_{\mathcal{X}} \int_{\mathcal{X}} \mu(x, x') \left\| x - x' \right\|^2 \mathrm{d}x \mathrm{d}x' \\ \text{subject to} \qquad B\mu = F\mu \\ B\mu = \nu_{\text{tgt}}. \end{aligned}$$

What is the justification for this objective though? The optimal policy might not be unique here, and there's no way to differentiate between "good" or "bad" ones. The objective only says "don't move around the states too much once you've hit the target", but doesn't say anything beyond that. (Also, this formulation doesn't say anything about the source distribution, which may or may not be a bad thing. If we want to connect with OT, it's a bad thing for sure.)

1.2. Lazy approach 2: discounted transients

The second most straightforward approach is to adjust the LP by introducing discounting:

$$\begin{aligned} &\min_{\mu} \int_{\mathcal{X}} \int_{\mathcal{X}} \mu(x, x') \left\|x - x'\right\|^2 \mathrm{d}x \mathrm{d}x' \\ \text{subject to} \qquad & B\mu = (1-\gamma)\nu_{\text{src}} + \gamma F\mu \\ & B\mu = (1-\gamma)\nu_{\text{src}} + \gamma\nu_{\text{tgt}}. \end{aligned}$$

The source is included in the second constraint to make sure that the problem is feasible. Unlike the previous problem formulation, this one is somewhat interpretable: I give you a geometric time horizon to move the source to the target with minimal total transport cost. So basically the dynamic OT problem but with a random time horizon. That helps because the optimal policy is stationary and the value function does not depend on the time index either; all the objects to learn are simpler. It's not clear if the cool properties of the classic dynamic OT solution would be inherited as nicely as in the fixed-horizon case... but that might just make things more interesting. Also, note that sending γ to 1 in the limit recovers the previous problem formulation.